



ASTRAQUANTUM TECH

SUMMER OF CYBERSECURITY 2026

WEEK 7 — SOC / BLUE TEAM ASSESSMENT

Logging • Monitoring • Log Analysis • Alert Investigation • Incident Response

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1. Executive Summary

This report presents the practical work completed for the SOC assessment. The activities focused on Security Operations Center concepts, alert generation and management, alert triage, structured log analysis, incident timeline development, and professional SOC case reporting.

The practical activities were completed using TryHackMe SOC Fundamentals, TryHackMe SOC L1 Alert Triage, and a free Blue Team Labs Online phishing investigation challenge. During the exercises, different security alerts were reviewed, relevant alert fields were examined, and alerts were prioritised and investigated using an L1 SOC workflow.

A Potential Data Exfiltration alert was selected for structured investigation and case reporting. The investigation included analysis of the available alert information, surrounding events, severity, status, and analyst actions. The report also documents the 5 Ws, incident timeline, evidence, impact, recommended response, and escalation decision.

2. Scope & Authorized Environment

The assessment was conducted within the authorized practical environments provided by the course.

The main platforms used were

- TryHackMe SOC Fundamentals

- TryHackMe SOC L1 Alert Triage

- Blue Team Labs Online (BTLO)

- TryHackMe SOC L1 Alert Triage SOC Dashboard

All investigations and activities were performed within the provided lab environments for educational and defensive security purposes. No unauthorized systems or real-world targets were intentionally tested.

3. Tools / Platforms Used

TryHackMe

TryHackMe was used to complete the SOC Fundamentals and SOC L1 Alert Triage learning activities. These rooms provided practical exercises involving SOC roles, alert properties, alert prioritisation, investigation, and alert lifecycle management.

TryHackMe SOC Dashboard

The SOC dashboard was used to review alerts, examine alert properties, assign alerts, change alert statuses, and perform the required alert-triage workflow.

Blue Team Labs Online

Blue Team Labs Online was used to complete the **Phishing Analysis** defensive investigation challenge. The investigation involved analysing a phishing email, examining email headers, identifying an originating IP address, investigating a suspicious URL, and reviewing the associated webpage.

Supporting Investigation Tools

The phishing investigation also used:

WHOIS / reverse DNS lookup

URL2PNG

Text editor for examining the .eml file

4. Learning Summary from Each Lab

4.1 TryHackMe — SOC Fundamentals

The SOC Fundamentals room introduced the role and purpose of a Security Operations Center. It covered the three SOC pillars: People, Process, and Technology. The room also introduced SOC L1, L2, and L3 responsibilities, SIEM, EDR, firewalls, alert triage, and the 5 Ws.

A practical port scan scenario was investigated using What, When, Where, Who, and Why. The activity was identified as an intended vulnerability assessment because the vulnerability assessment team had informed the SOC about the scan.

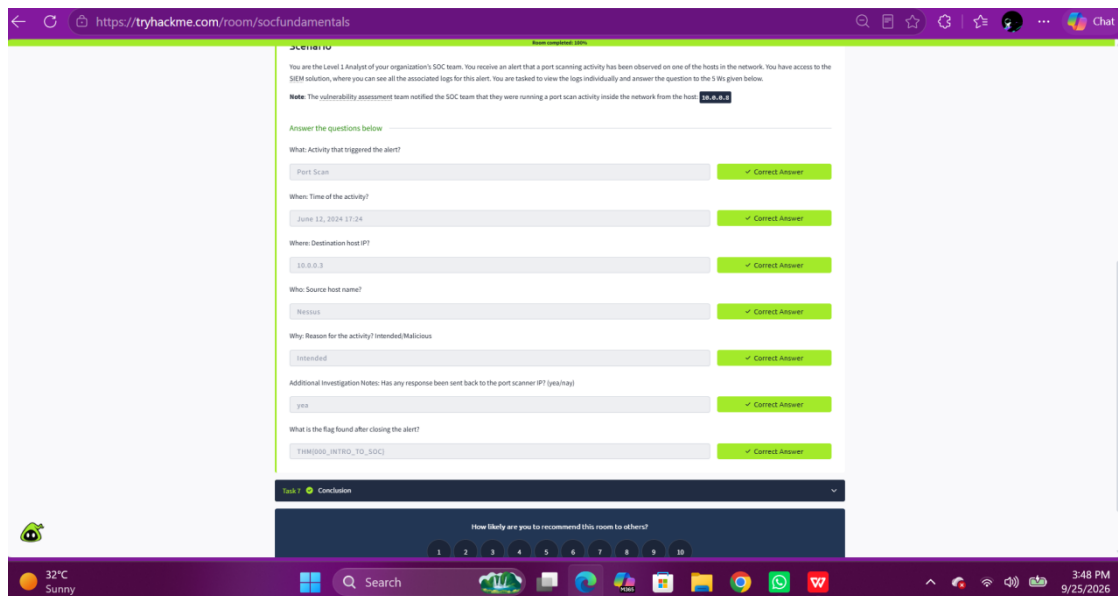


Figure 1: TryHackMe SOC Fundamentals practical exercise showing the completed 5 Ws investigation.

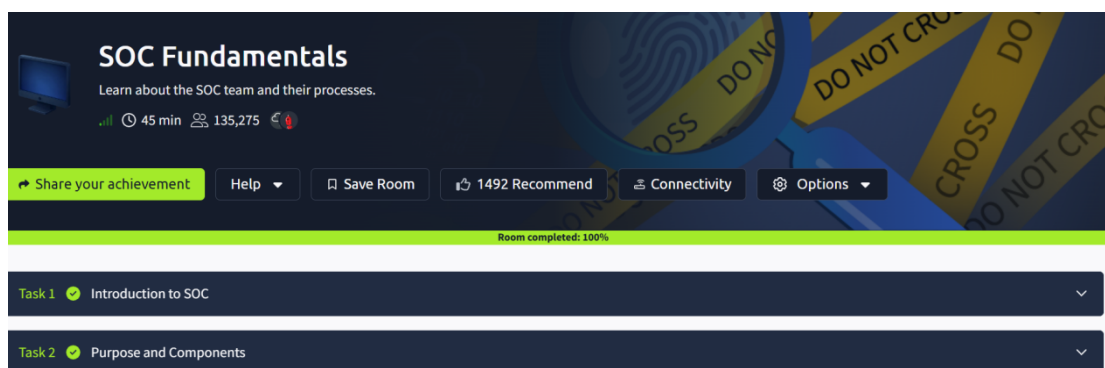


Figure 2: Successful completion of the TryHackMe SOC Fundamentals room.

4.2 TryHackMe — SOC L1 Alert Triage

The SOC L1 Alert Triage room focused on the alert lifecycle from event generation to investigation and resolution. It covered alert fields, severity, status, verdict, assignee, alert prioritisation, and the responsibilities of an L1 analyst.

The SOC dashboard initially contained five alerts. Alerts were prioritised according to severity and age. The first-priority alert was **Potential Data Exfiltration**, which was assigned to the analyst and changed to **In Progress** for investigation.

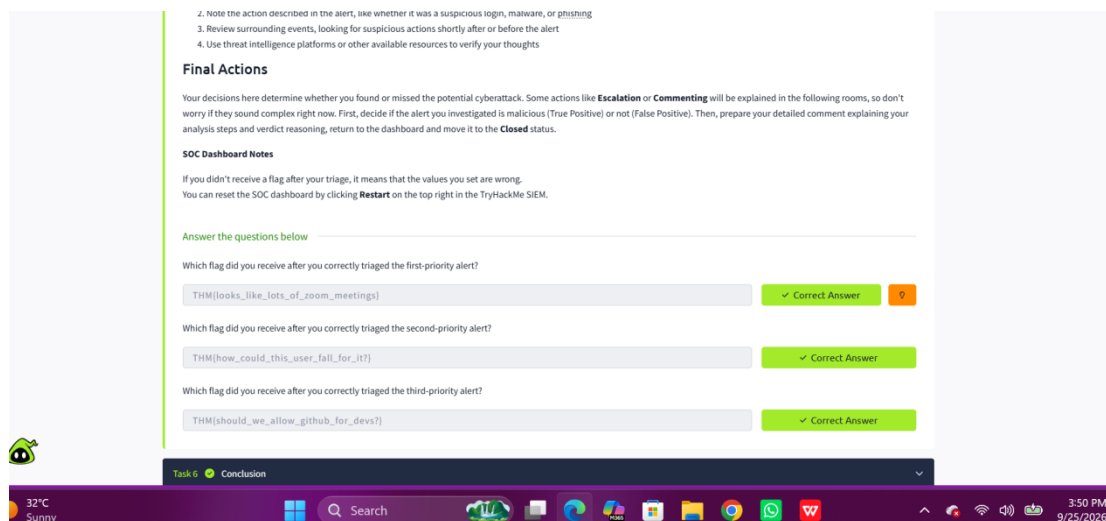


Figure 3: TryHackMe SOC L1 Alert Triage dashboard showing the initial alert queue and alert properties.

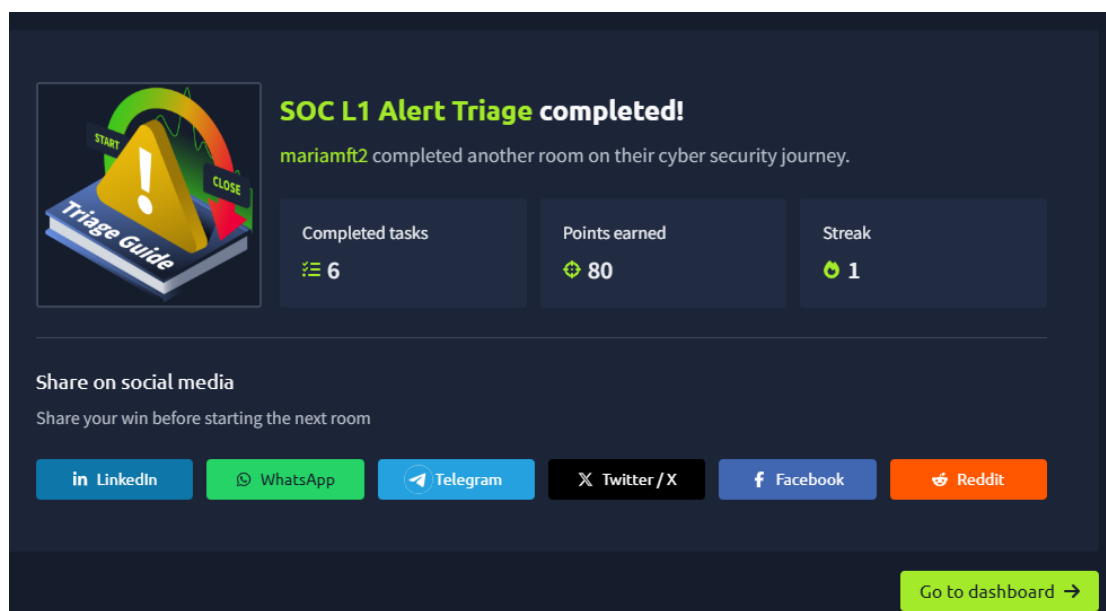


Figure 4: Successful completion of the TryHackMe SOC L1 Alert Triage practical activity.

4.3 Blue Team Labs Online — Phishing Analysis

The Phishing Analysis challenge involved investigating a phishing-related email and identifying useful forensic artifacts.

The investigation identified.

Suspicious Blogspot URL

Originating IP address 103.9.171.10

Email attachment/file Website Content Form Submission.eml

Reverse DNS/WHOIS information

A webpage that displayed “**Blog has been removed**”

The investigation concluded that the email contained suspicious web content associated with phishing activity.

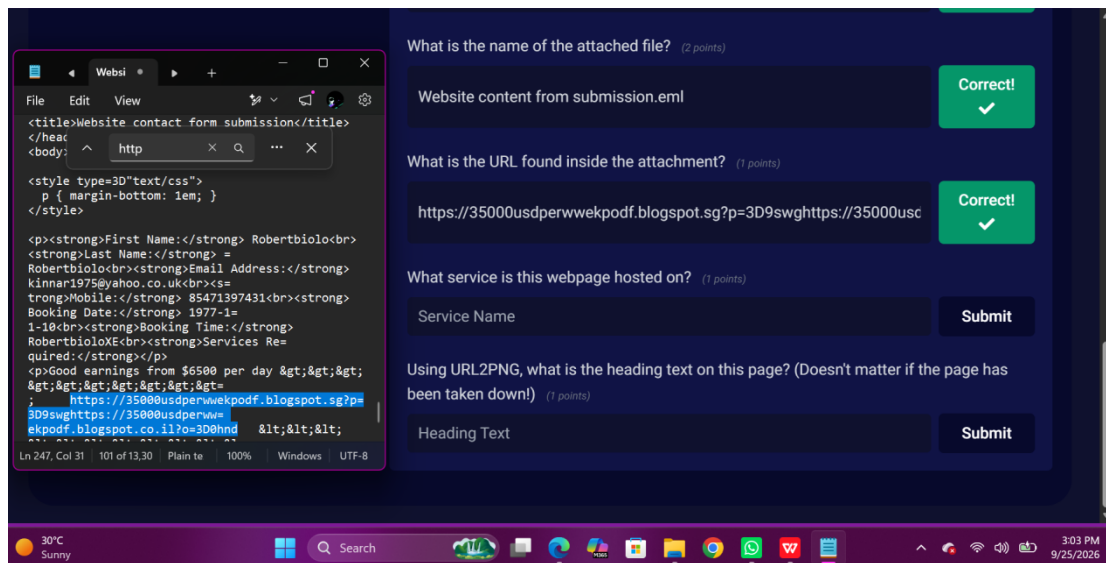


Figure 5: Phishing email and suspicious URL identified during the investigation.

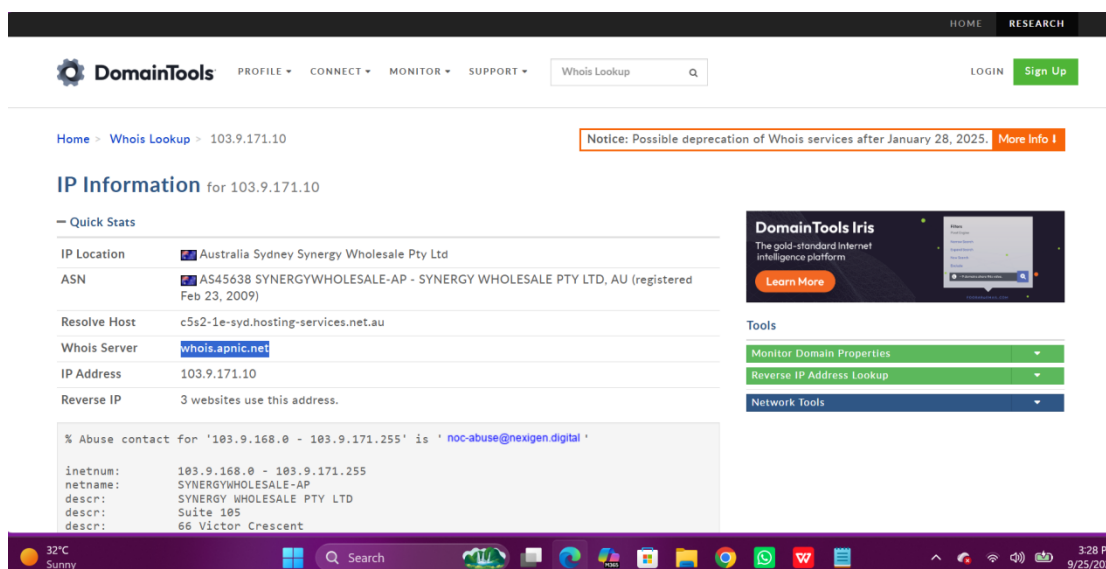


Figure 6: WHOIS and reverse DNS information for the identified originating IP address.

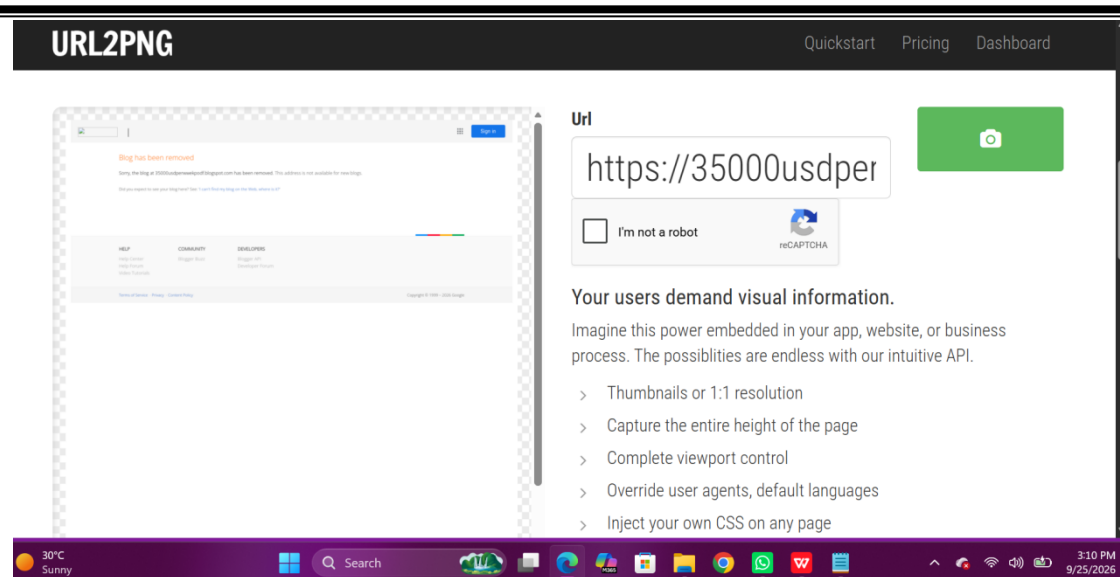


Figure 7: URL2PNG capture showing that the associated Blogspot page had been removed.



Figure 8: Successful completion of the BTLO Phishing Analysis challenge.

5. Alert Investigation

Selected Alert

Alert Name: Potential Data Exfiltration

Alert Time: March 21, 2025 at 13:30

Severity: Critical

Initial Status: Awaiting Action

Initial Assignee: None

Initial Verdict: None

The Potential Data Exfiltration alert was selected for investigation according to the alert prioritisation process. The alert was assigned to the L1 analyst and its status was changed to **In Progress**.

6. 5 Ws Analysis

What?

A Potential Data Exfiltration alert was generated, indicating potentially suspicious data-transfer activity.

Who?

The specific user responsible for the activity was not identified in the available dashboard evidence.

Where?

The affected host, IP address, or asset was not identified in the available dashboard evidence.

When?

The alert was recorded on **March 21, 2025 at 13:30**.

Why?

The alert was generated because the security monitoring system detected activity matching the detection logic for potential data exfiltration.

7. Incident Timeline

Time	Event	Evidence Source	Why It Matters
March 21, 2025 — 11:53	Brute Force Attack from External alert was recorded.	TryHackMe SOC L1 Alert Triage dashboard	Provides earlier security-event context.
March 21, 2025 — 12:40	Unusual VPN Login Location alert was recorded.	TryHackMe SOC L1 Alert Triage dashboard	Shows an alert that was later classified as False Positive.
March 21, 2025 — 13:02	Download from GitHub Repository alert was recorded.	TryHackMe SOC L1 Alert Triage dashboard	Provides surrounding context before the selected alert.
March 21, 2025 — 13:30	Potential Data Exfiltration alert was generated with Critical severity.	TryHackMe SOC L1 Alert Triage dashboard	Identifies the main alert selected for investigation.
March 21, 2025 — 13:58	Double-Extension File Creation alert was recorded.	TryHackMe SOC L1 Alert Triage dashboard	Provides a subsequent security event.
After alert selection	Potential Data Exfiltration was assigned to the L1 analyst and changed to In Progress.	TryHackMe SOC L1 Alert Triage dashboard	Shows that active investigation had started.

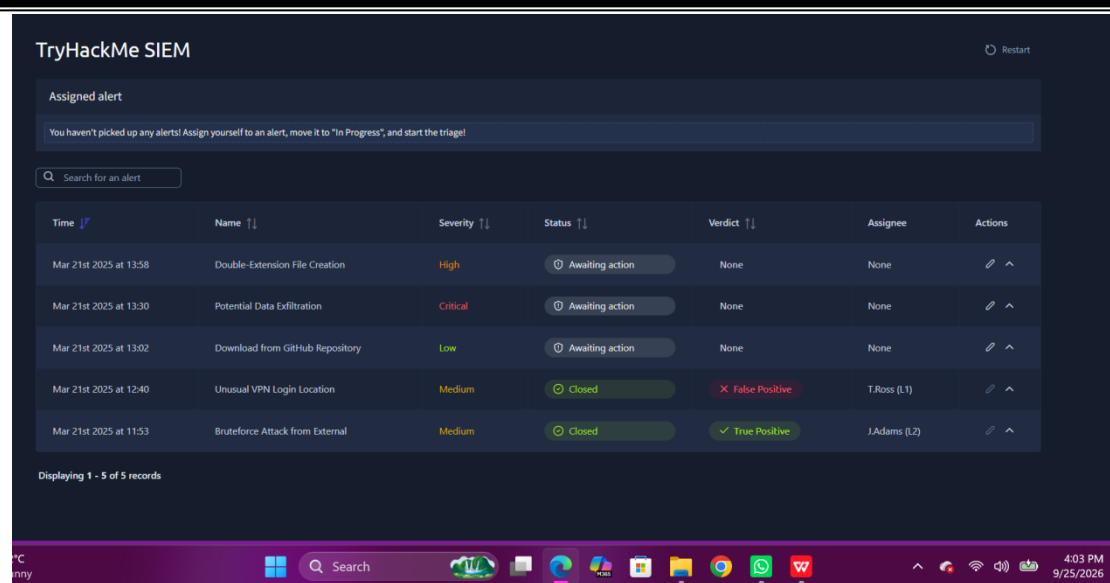


Figure 9: Initial SOC L1 Alert Triage dashboard showing the chronological alert queue.

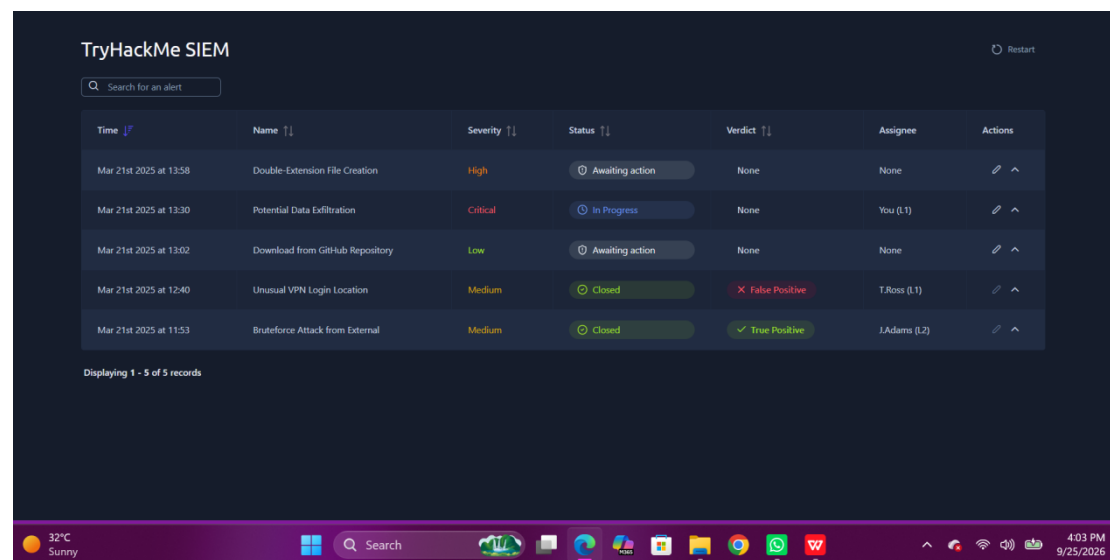


Figure 10: Potential Data Exfiltration alert after assignment to the L1 analyst and status change to In Progress.

8. Evidence / Screenshots

The investigation evidence consisted of the alert properties and dashboard information available during the practical exercise.

Important evidence included:

Alert name

Alert timestamp

Critical severity

Initial Awaiting Action status

Initial absence of an assignee

Surrounding alerts

Assignment of the alert to the L1 analyst

Status change to In Progress

Additional phishing investigation evidence was documented in the BTLO section of this report.

9. Verdict

Verdict: Needs Escalation

The Potential Data Exfiltration alert was classified as **Needs Escalation** because it had Critical severity and the available dashboard evidence was not sufficient to determine whether the underlying activity was a genuine unauthorized data-exfiltration incident or legitimate activity.

Further investigation by SOC L2 is required before making a final True Positive or False Positive determination.

10. Severity

Critical

The alert was displayed with Critical severity in the SOC L1 Alert Triage dashboard. Critical severity indicates that the alert requires urgent investigation according to the prioritisation process.

11. Impact

Potential data exfiltration could involve unauthorized transfer of sensitive organizational information. However, the actual impact cannot be confirmed from the available dashboard evidence because the affected data, user, destination, and amount of transferred information were not provided.

12. Recommended Response

The alert should be escalated to SOC L2 for deeper investigation.

The investigation should identify the affected user and host, review relevant SIEM/EDR logs, examine network connections and data-transfer activity, and determine whether sensitive information was accessed or transferred.

If unauthorized exfiltration is confirmed, containment actions should be performed according to the organization's incident-response procedures.

13. Conclusion

This assessment provided practical experience with fundamental SOC operations, including security monitoring, alert management, alert prioritisation, alert triage, evidence analysis, and incident reporting.

The practical exercises demonstrated the importance of reviewing alert properties, investigating surrounding activity, and using evidence before assigning a final verdict. The SOC L1 workflow showed how analysts take ownership of alerts, investigate suspicious activity, document their findings, and escalate cases when additional investigation is required.

The structured investigation of the Potential Data Exfiltration alert demonstrated the importance of severity, timestamps, surrounding events, evidence, and escalation decisions when handling security alerts.

14. Appendix

Appendix A — Assessment Questions

Q24. Differentiate between a raw event, alert, and incident.

Raw Event: An individual activity recorded by a system, such as a login, file download, or process execution.

Alert: A notification generated by a security solution when an event or sequence of events matches a detection rule.

Incident: A confirmed or suspected security event that requires investigation and potentially a response.

Q25. Why are timestamps important when correlating logs?

Timestamps allow analysts to place events in chronological order and correlate activity occurring across different systems. Accurate timestamps help establish what happened before, during, and after an alert.

Q26. Why can one log entry be misleading without surrounding events?

A single log entry may not provide enough context to determine whether activity is malicious or legitimate. Reviewing surrounding events helps analysts understand the complete sequence of activity.

Q27. What is a false positive? Give a SOC example.

A false positive occurs when a security system generates an alert for activity that is not actually a security threat.

For example, an unusual VPN login location may trigger an alert even though the user was legitimately travelling or using an approved connection.

Q28. What is the purpose of a SIEM?

A SIEM collects and centralises logs from different systems, applies detection and correlation rules, and generates alerts that help SOC analysts identify suspicious activity.

Q29. When should an L1 analyst escalate instead of closing an alert?

An L1 analyst should escalate when the activity appears to be a genuine threat, when the alert has significant severity, or when the available evidence is insufficient to confidently determine whether the activity is malicious or legitimate.

Q30. What evidence should be collected before recommending containment?

Relevant evidence can include affected hosts and users, timestamps, source and destination IP addresses, processes, files, authentication events, network connections, related alerts, and surrounding SIEM/EDR logs.